

Foundation (Jalapeno)

Q1. Factor the expression: $x^2 - 4$

- (A) $x^2 - 2^2$
- (B) $(x + 2)(x - 2)$
- (C) $(x - 2)(x + 2)$
- (D) $x^2 - 2$
- (E) $2x^2 - 2$

Q2. What is the factored form of $a^2 - 9$?

- (A) $a^2 - 3^2$
- (B) $(a - 3)(a + 3)$
- (C) $(a + 3)(a - 3)$
- (D) $a^2 - 3$
- (E) $3a^2 - 3$

Q3. Factor the expression: $y^2 - 25$

- (A) $y^2 - 5^2$
- (B) $(y + 5)(y - 5)$
- (C) $(y - 5)(y + 5)$
- (D) $y^2 - 5$
- (E) $5y^2 - 5$

Q4. Which of the following is a difference of squares?

- (A) $x^2 + 2$
- (B) $x^2 - 2$
- (C) $x^2 + 3x$
- (D) $x^2 - 4$
- (E) $2x^2 - 2$

Q5. What is the factored form of $16 - x^2$?

- (A) $4^2 - x^2$
- (B) $(4 - x)(4 + x)$
- (C) $(x - 4)(x + 4)$
- (D) $x^2 - 4$
- (E) $4x^2 - 4$

Q6. Factor the expression: $36 - y^2$

- (A) $6^2 - y^2$
- (B) $(y + 6)(y - 6)$
- (C) $(y - 6)(y + 6)$
- (D) $y^2 - 6$
- (E) $6y^2 - 6$

Q7. What is the factored form of $1 - x^2$?

- (A) $1^2 - x^2$
- (B) $(x + 1)(x - 1)$
- (C) $(x - 1)(x + 1)$
- (D) $x^2 - 1$
- (E) $x^2 - x$

Q8. Factor the expression: $x^2 - 49$

- (A) $x^2 - 7^2$
- (B) $(x + 7)(x - 7)$
- (C) $(x - 7)(x + 7)$
- (D) $x^2 - 7$
- (E) $7x^2 - 7$

Open Ended Questions (FRQ)

Q9. Factor the expression: $a^2 - b^2$

Q10. Use the difference of squares to factor the expression: $(2y)^2 - (3x)^2$ and simplify.

Chef's Tips

- Remind students to look for perfect square binomials.
- Encourage students to check their solutions by multiplying the factors.
- Have students work in pairs to simplify and check their work on selected problems.